Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2025	ENGI3411-WE01

Title: (BEng/MEng)
Solid Mechanics 3

Time Allowed:	2 hours
Additional Material provided:	Yes – formula sheet included as part of the exam paper.
Materials permitted:	N/A
Calculators Permitted:	Those from the Casio fx-83 and fx-85 series.

Instructions to Candidates:	This is an in-person, invigilated exam. Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

Q1

You are responsible for designing the cylinders of a Formula 1 car. Cylinders are thin-walled components under extremely high, oscillating internal pressure, as shown in Figure Q1-1. Cylinder walls tolerate this pressure as hoop stress calculated using $(\sigma_h = \frac{pr}{t})$, where p is the internal pressure, r is the cylinder radius, and t is the thickness of the cylinder wall. These cylinders are made of an aluminium alloy with the yield stress of $480\,\text{MPa}$, and fracture toughness of $18\,\text{MPa}\sqrt{\text{m}}$.

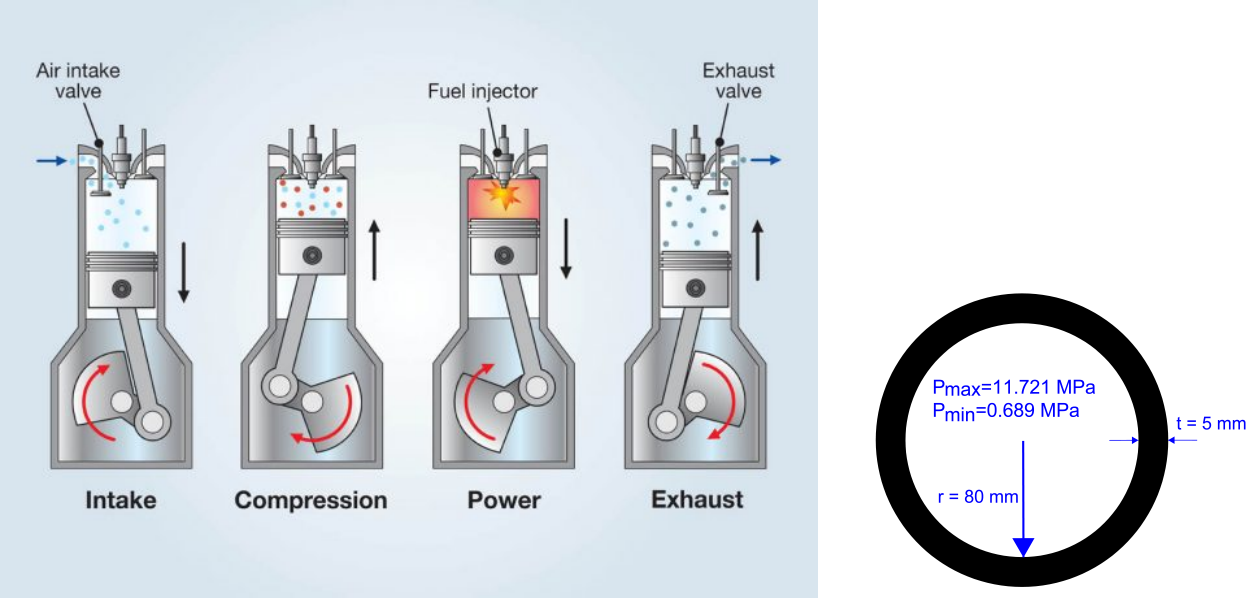


Figure Q1-1: The schematic view of cylinders and the Hoop stress.

(a) Based on your engineering judgment, should you design this system using the leak-before-break principle? Justify your response in a few sentences. **[15%]**

(b) The thickness of the cylinders is $t = 5\,\text{mm}$, their **internal** radius is $r = 80\,\text{mm}$ and height is $h = 60\,\text{cm}$. Several tests in your company show that the internal pressure of the cylinders reaches $11.721\,\text{MPa}$. Determine the critical crack length on the thin walls of the cylinders and discuss your assumptions for calculations. Use Figure Q1-2 diagram to determine the geometry factor (Y) if needed. **[40%]**

Cylinders are under cyclic load, where the maximum stress of $11.721\,\text{MPa}$ is applied while the pistons are at their highest position and the minimum stress of $0.689\,\text{MPa}$ is applied when the pistons are at their lowest position. The engine's revolutions per minute (RPM) determine the number of cycles that the cylinders go through in every minute of operation. While racing, the engine operates at an average of $12000\,\text{RPM}$ and races usually last for **Four** hours.

(c) The car engine is tested before every racing competition to detect cracks in its cylinders' walls. Use the Paris Law to determine the maximum crack length on the cylinder walls, which can be safely tolerated for a four-hour race. The Paris Law coefficients for the Aluminium alloy are $C = 10^{-9}$ and $m = 3.5$. You may consider the value of the Y constant throughout the crack growth equal to the safest engineering assumption you can make. Explain and justify your assumption. **[45%]**

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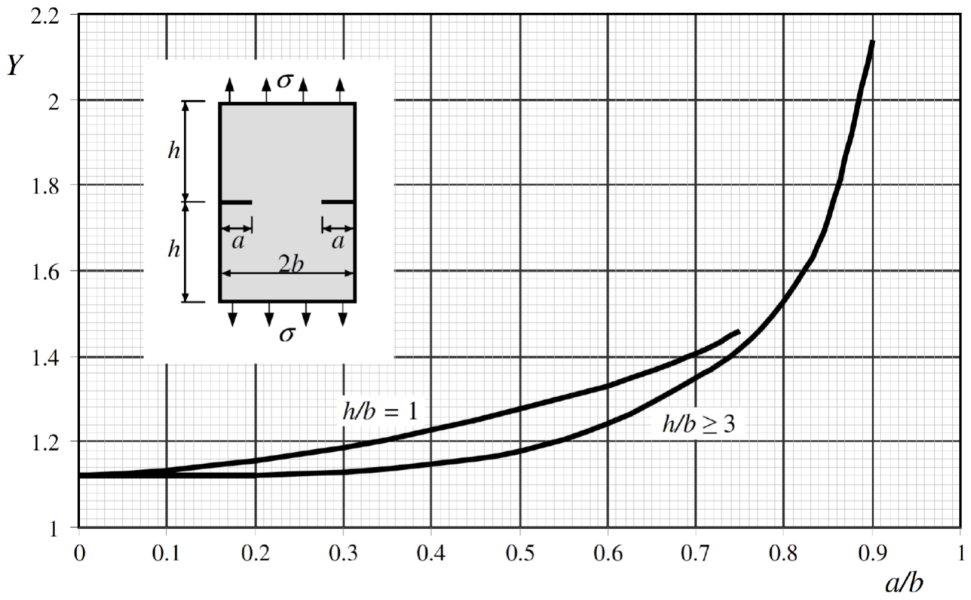


Figure Q1-2 Graph of Y factor for edge cracks in Aluminium alloy.

Q2

(a) Consider the four types of quadrilateral elements illustrated in Figure Q2-1 (a)-(d) along with the Pascal triangle of monomials shown in Figure Q2-1 (e). Write the general polynomial form for the shape functions N_i for each of the four quadrilateral elements? **[20%]**

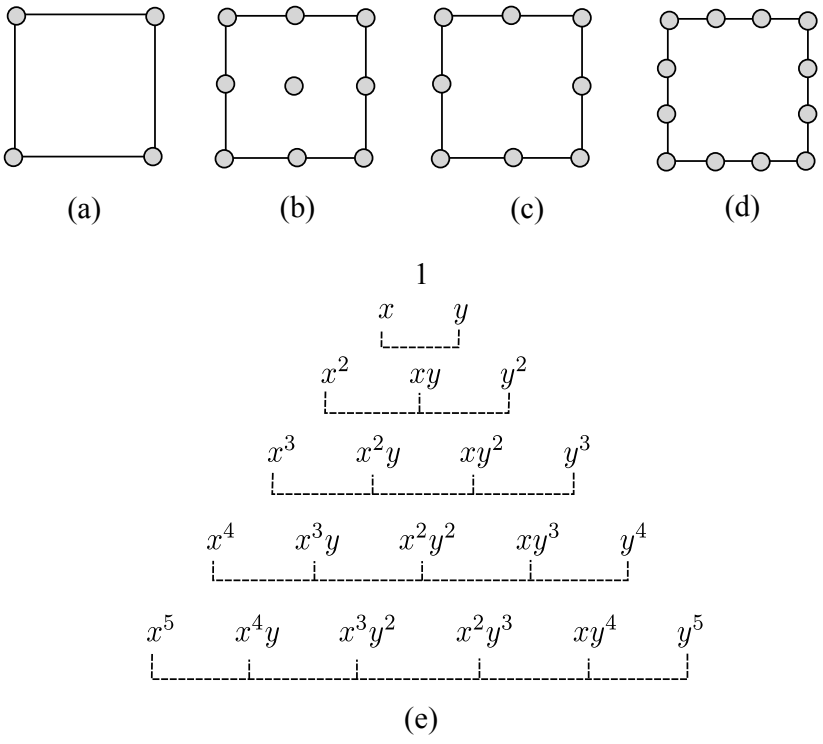


Figure Q2-1: Quadrilateral elements: (a) 4-node (b) 9-node (c) 8-node (d) 12-node, and (e) Pascal triangle of monomials.

(b) Consider the single triangular element shown in Figure Q2-2(a) in the global coordinates system (x, y) with coordinates given in mm. Assume plane stress conditions. The reference triangular element in the local coordinates (ξ, η), along with the associated shape functions, is also given in Figure Q2.2(b). Let $E = 210$ GPa and $\nu = 0.25$.

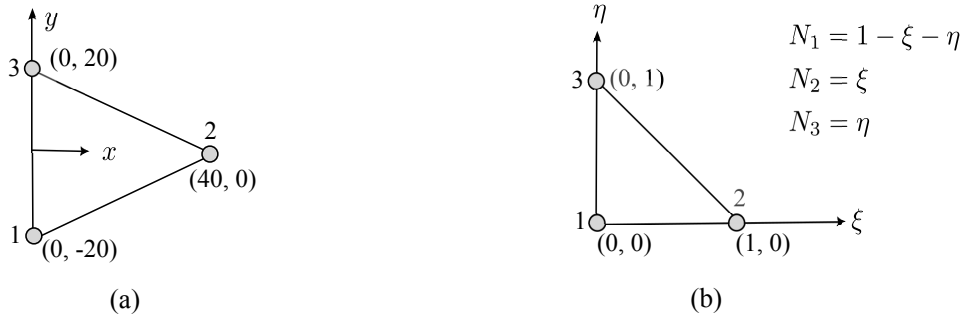


Figure Q2-2: (a) Triangular element in global coordinates, and (b) Triangular element in local coordinates with associated shape functions.

i. Using equations provided on the formula sheet, determine the full strain-displacement matrix $[B]$ for the element shown in Figure Q2-2(a). **[40%]**

ii. Without performing calculations, summaries the steps to calculate stiffness matrix $[K^e]$ for this element. Discuss the mapping between global and local coordinates, and also elaborate on the choice of the order of Gauss quadrature for the numerical integration required in the calculation of the $[K^e]$ matrix for this element. **[20%]**

iii. Assume the elemental nodal displacement have been determined to be $u_1 = 0.0$, $v_1 = 0.05$ mm, $u_2 = 0.025$ mm, $v_2 = 0.0$, $u_3 = 0.0$, $v_3 = 0.05$ mm. Determine the stress components in the element. **[20%]**

Q3

(a) Consider the following equation for the calculation of stiffness matrix using a Gauss quadrature

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T [D^e][B]_i w_i \det([J]_i).$$

Write down the size of the matrices in the element stiffness equation for an eight-noded quadrilateral element in axi-symmetric analysis. **[10%]**

(b) Consider the 4-node quadrilateral element in the global coordinate system, as shown in Figure Q3(a). The displacement boundary conditions and loading are also provided in the same figure. The quadrilateral element in the local coordinates along with its associated shape functions, are given in Figure Q3 (b). Assume plane stress condition with a thickness of $t = 0.1$m, Young's modulus $E = 2000$ Pa, and Poisson's ratio $\nu = 0.3$.

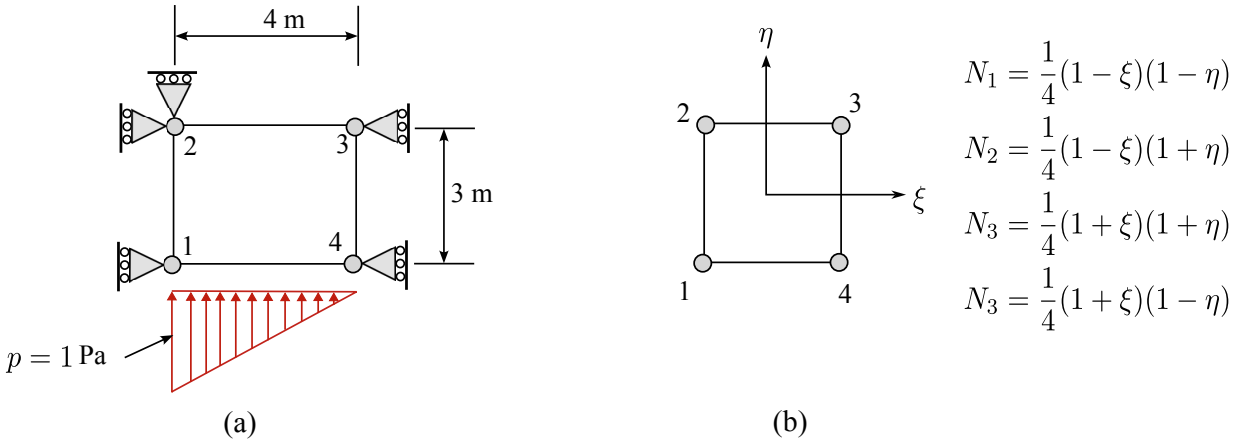


Figure Q3: (a) Four-node quadrilateral element in global coordinates system, and (b) four-noded quadrilateral element in local coordinates system with the associated shape functions.

i. Use the surface integral provided in the formula sheet to show that the equivalent nodal force vector $\{f^e\}$ associated with the traction load shown in Figure Q3 (a) is:

$$\{f^e\} = \begin{Bmatrix} 0 \\ 0.1333 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0.0667 \end{Bmatrix} \text{N}.$$

[40%]

ii. Consider the following stiffness matrix for the element shown in Figure Q3 (a)

$$[K^e] = \begin{bmatrix} 89 & 36 & -7 & 3 & -45 & -36 & -38 & -3 \\ 36 & 117 & -3 & -88 & -36 & -58 & 3 & 30 \\ -7 & -3 & 89 & -36 & -38 & 3 & -45 & 36 \\ 3 & -88 & -36 & 117 & -3 & 30 & 36 & -58 \\ -45 & -36 & -38 & -3 & 89 & 36 & -7 & 3 \\ -36 & -58 & 3 & 30 & 36 & 117 & -3 & -88 \\ -38 & 3 & -45 & 36 & -7 & -3 & 89 & -36 \\ -3 & 30 & 36 & -58 & 3 & -88 & -36 & 117 \end{bmatrix} \text{ N/m.}$$

Find the reduced stiffness matrix and the associated load vector according to the boundary conditions shown in Figure Q3 (a). **[20%]**

iii. Solve the resulting system of equations from the previous part to calculate the unknown nodal displacements. **[30%]**

ENGI3411-WE01 Solid Mechanics 3
Course: Fracture and Fatigue
Formula sheet

Stress and displacement distribution locally around the tip of a mode I crack oriented in the x-direction

$$\sigma_{xx} = \frac{K_I}{\sqrt{2\pi r}}\cos\frac{\theta}{2}\left[1-\sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]+\ldots$$

$$\sigma_{yy} = \frac{K_I}{\sqrt{2\pi r}}\cos\frac{\theta}{2}\left[1+\sin\frac{\theta}{2}\sin\frac{3\theta}{2}\right]+\ldots$$

$$\sigma_{xy} = \frac{K_I}{\sqrt{2\pi r}}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\cos\frac{3\theta}{2}+\ldots$$

$$u_x = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\cos\frac{\theta}{2}\left[\frac{1}{2}(\kappa-1)+\sin^2\frac{\theta}{2}\right]+\cdots$$

$$u_y = \frac{K_\mathrm{I}}{\mu}\sqrt{\frac{r}{2\pi}}\sin\frac{\theta}{2}\left[\frac{1}{2}(\kappa+1)-\cos^2\frac{\theta}{2}\right]+\cdots$$

$\kappa = 3-4\nu$ (plane strain), or $(3-\nu)/(1+\nu)$ (plane stress)

Mode I stress intensity factor

$$K_I = Y\sigma\sqrt{\pi a}$$

Internal crack of length $2a$ in infinite plate: $Y = 1$
Edge crack of length a in relatively large plate: $Y = 1.12$

Plastic zone correction

$$\text{Equivalent crack length} = a + \frac{K_I^2}{2\pi\sigma_Y^2}$$

Failure diagrams for mean stress correction

Modified Goodman:

$$S_a = S_D\left[1-\frac{S_{mean}}{S_u}\right]$$

Gerber parabola:

$$S_a = S_D\left[1-\left(\frac{S_{mean}}{S_u}\right)^2\right]$$

Soderberg:

$$S_a = S_D\left[1-\frac{S_{mean}}{S_{yield}}\right]$$

Strain-life method

Strain life equation:

$$\frac{\Delta\varepsilon}{2} = \frac{\sigma'_f}{E}(2N_f)^b + \varepsilon'_f(2N_f)^c$$

Mean stress correction (Smith Watson Topper):

$$\left(\sigma_{mean}+\frac{\Delta\sigma}{2}\right)\frac{\Delta\varepsilon}{2} = \frac{\sigma'^2_f}{E}(2N_f)^{2b} + \sigma'_f\varepsilon'_f(2N_f)^{b+c}$$

continued

Macro crack growth laws based on fracture mechanics approach

Paris Law:

$$\frac{da}{dN} = C(\Delta K)^m$$

Broek and Schijve:

$$\frac{da}{dN} = CK_{max}^2 \Delta K$$

Walker:

$$\frac{da}{dN} = CK_{max}^m (\Delta K)^n$$

Walker:

$$\frac{da}{dN} = C\overline{\Delta K}^n \qquad \text{where } \overline{\Delta K} = S_{max}(1-R)^m \sqrt{\pi a}$$

Forman:

$$\frac{da}{dN} = \frac{C(\Delta K)^m}{(1-R)(K_{Ic} - K_{max})}$$

continued

ENGI3411-WE01 Solid Mechanics 3
Course: Finite-element method
Formula sheet

Three-dimensional stress-strain relationship

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{xy} \\ \sigma_{yz} \\ \sigma_{zx} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & \nu & 0 & 0 & 0 \\ \nu & (1-\nu) & \nu & 0 & 0 & 0 \\ \nu & \nu & (1-\nu) & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix}$$

Two-dimensional stress-strain relationships

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} (1-\nu) & \nu & 0 \\ \nu & (1-\nu) & 0 \\ 0 & 0 & \frac{(1-2\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \quad \text{(plane strain)}$$

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{(1-\nu)}{2} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{Bmatrix} \quad \text{(plane stress)}$$

Three-dimensional strain definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = [L]\{u\} = \begin{bmatrix} \frac{\partial}{\partial x} & 0 & 0 \\ 0 & \frac{\partial}{\partial y} & 0 \\ 0 & 0 & \frac{\partial}{\partial z} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial z} & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} & 0 & \frac{\partial}{\partial x} \end{bmatrix} \begin{Bmatrix} u \\ v \\ w \end{Bmatrix}$$

Interpolation using shape functions

$$u = \sum_{i=1}^{n_{en}} N_i u_i = \{N\}^T \{u\}$$

Three-dimensional strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial x} & 0 & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial y} & 0 \\ 0 & 0 & \frac{\partial N_1}{\partial z} & \dots & 0 & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & 0 & \dots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial y} & \dots & 0 & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial z} & 0 & \frac{\partial N_1}{\partial x} & \dots & \frac{\partial N_{n_{en}}}{\partial z} & 0 & \frac{\partial N_{n_{en}}}{\partial x} \end{bmatrix}}_{[B]=[L][N]} \begin{Bmatrix} u_1 \\ v_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}$$

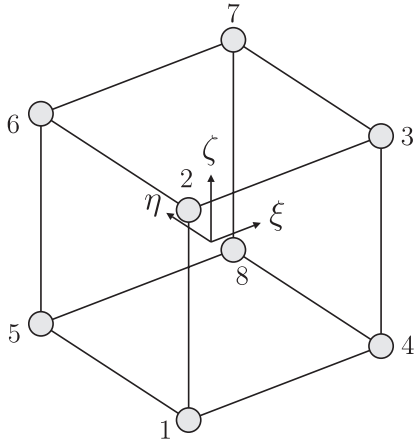
continued

Plane-stress/strain strain-displacement definition

$$\left\{ \begin{array}{c} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{array} \right\} = \left[\begin{array}{ccccc} \frac{\partial N_1}{\partial x} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \cdots & \frac{\partial N_{n_{en}}}{\partial y} & \frac{\partial N_{n_{en}}}{\partial x} \end{array} \right] \left\{ \begin{array}{c} u_1 \\ v_1 \\ \vdots \\ u_{n_{en}} \\ v_{n_{en}} \end{array} \right\}.$$

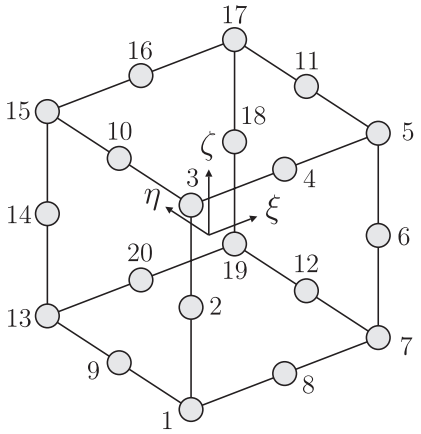
Shape functions for hexahedral elements

Tri-linear eight-noded hexahedral



$$\begin{aligned} N_1 &= (1-\xi)(1-\eta)(1-\zeta)/8 \\ N_2 &= (1-\xi)(1-\eta)(1+\zeta)/8 \\ N_3 &= (1+\xi)(1-\eta)(1+\zeta)/8 \\ N_4 &= (1+\xi)(1-\eta)(1-\zeta)/8 \\ N_5 &= (1-\xi)(1+\eta)(1-\zeta)/8 \\ N_6 &= (1-\xi)(1+\eta)(1+\zeta)/8 \\ N_7 &= (1+\xi)(1+\eta)(1+\zeta)/8 \\ N_8 &= (1+\xi)(1+\eta)(1-\zeta)/8 \end{aligned}$$

Tri-quadratic twenty-noded hexahedral



$$\begin{aligned} N_1 &= (1-\xi)(1-\eta)(1-\zeta)(-\xi-\eta-\zeta-2)/8 \\ N_2 &= (1-\xi)(1-\eta)(1-\zeta^2)/4 \\ N_3 &= (1-\xi)(1-\eta)(1+\zeta)(-\xi-\eta+\zeta-2)/8 \\ N_4 &= (1-\xi^2)(1-\eta)(1+\zeta)/4 \\ N_5 &= (1+\xi)(1-\eta)(1+\zeta)(\xi-\eta+\zeta-2)/8 \\ N_6 &= (1+\xi)(1-\eta)(1-\zeta^2)/4 \\ N_7 &= (1+\xi)(1-\eta)(1-\zeta)(\xi-\eta-\zeta-2)/8 \\ N_8 &= (1-\xi^2)(1-\eta)(1-\zeta)/4 \\ N_9 &= (1-\xi)(1-\eta^2)(1-\zeta)/4 \\ N_{10} &= (1-\xi)(1-\eta^2)(1+\zeta)/4 \\ N_{11} &= (1+\xi)(1-\eta^2)(1+\zeta)/4 \\ N_{12} &= (1+\xi)(1-\eta^2)(1-\zeta)/4 \\ N_{13} &= (1-\xi)(1+\eta)(1-\zeta)(-\xi+\eta-\zeta-2)/8 \\ N_{14} &= (1-\xi)(1+\eta)(1-\zeta^2)/4 \\ N_{15} &= (1-\xi)(1+\eta)(1+\zeta)(-\xi+\eta+\zeta-2)/8 \\ N_{16} &= (1-\xi^2)(1+\eta)(1+\zeta)/4 \\ N_{17} &= (1+\xi)(1+\eta)(1+\zeta)(\xi+\eta+\zeta-2)/8 \\ N_{18} &= (1+\xi)(1+\eta)(1-\zeta^2)/4 \\ N_{19} &= (1+\xi)(1+\eta)(1-\zeta)(\xi+\eta-\zeta-2)/8 \\ N_{20} &= (1-\xi^2)(1+\eta)(1-\zeta)/4 \end{aligned}$$

continued

Axi-symmetric strain-displacement definition

$$\begin{Bmatrix} \varepsilon_{rr} \\ \varepsilon_{zz} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{Bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial r} & 0 & \cdots & \frac{\partial N_{n_{en}}}{\partial r} & 0 \\ 0 & \frac{\partial N_1}{\partial z} & \cdots & 0 & \frac{\partial N_{n_{en}}}{\partial z} \\ \frac{N_1}{r_1} & 0 & \cdots & \frac{N_{n_{en}}}{r_{n_{en}}} & 0 \\ \frac{\partial N_1}{\partial z} & \frac{\partial N_1}{\partial r} & \cdots & \frac{\partial N_{n_{en}}}{\partial z} & \frac{\partial N_{n_{en}}}{\partial r} \end{bmatrix} \begin{Bmatrix} u_1 \\ w_1 \\ \vdots \\ u_{n_{en}} \\ w_{n_{en}} \end{Bmatrix}.$$

Global shape function derivatives

$$\begin{Bmatrix} \frac{\partial N_i}{\partial x} \\ \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial z} \end{Bmatrix} = \begin{bmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \eta}{\partial x} & \frac{\partial \zeta}{\partial x} \\ \frac{\partial \xi}{\partial y} & \frac{\partial \eta}{\partial y} & \frac{\partial \zeta}{\partial y} \\ \frac{\partial \xi}{\partial z} & \frac{\partial \eta}{\partial z} & \frac{\partial \zeta}{\partial z} \end{bmatrix} \begin{Bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \\ \frac{\partial N_i}{\partial \zeta} \end{Bmatrix} = [J]^{-1} \begin{Bmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \\ \frac{\partial N_i}{\partial \zeta} \end{Bmatrix}.$$

The Jacobian matrix

$$[J] = \begin{bmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} & \frac{\partial z}{\partial \xi} \\ \frac{\partial x}{\partial \eta} & \frac{\partial y}{\partial \eta} & \frac{\partial z}{\partial \eta} \\ \frac{\partial x}{\partial \zeta} & \frac{\partial y}{\partial \zeta} & \frac{\partial z}{\partial \zeta} \end{bmatrix} = \begin{bmatrix} \frac{\partial N_1}{\partial \xi} & \frac{\partial N_2}{\partial \xi} & \cdots & \frac{\partial N_{n_{en}}}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} & \frac{\partial N_2}{\partial \eta} & \cdots & \frac{\partial N_{n_{en}}}{\partial \eta} \\ \frac{\partial N_1}{\partial \zeta} & \frac{\partial N_2}{\partial \zeta} & \cdots & \frac{\partial N_{n_{en}}}{\partial \zeta} \end{bmatrix} \begin{bmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ \vdots & \vdots & \vdots \\ x_{n_{en}} & y_{n_{en}} & z_{n_{en}} \end{bmatrix}$$

Weak form of equilibrium

$$\int_V [B]^T [D^{\mathrm{e}}][B] dV \{d\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

Element stiffness and force vector

$$[k^e] = \int_V [B]^T [D^{\mathrm{e}}][B] dV \qquad \text{and} \qquad \{f^e\} = \int_S [N]^T \{t\} dS + \int_V [N]^T \{f^b\} dV.$$

Gauss-Legendre quadrature

$$\int_{-1}^{1} f(x) dx \approx \sum_{i=1}^{n_{Gp}} f(x_i) w_i$$

n_{Gp}	Abscissae, x_i	Weights, w_i
1	0	2
2	$\pm 1/\sqrt{3}$	1
3	0	$8/9$
	$\pm\sqrt{3/5}$	$5/9$
4	$\pm\sqrt{\frac{3}{7} - \frac{2}{7}\sqrt{\frac{6}{5}}}$	$(18+\sqrt{30})/36$
	$\pm\sqrt{\frac{3}{7} + \frac{2}{7}\sqrt{\frac{6}{5}}}$	$(18-\sqrt{30})/36$

Numerical approximation of the element stiffness matrix

$$[k^e] \approx \sum_{i=1}^{n_{Gp}} [B]_i^T [D^{\mathrm{e}}][B]_i w_i \det([J]_i)$$

The inverse of a 2×2 matrix is calculated using:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}, \quad \textit{where } \det(A) = ad - bc$$

END